head; you are being stretched like a piece of taffy candy, but gently. The cause, you realize, is the hole's gravity: Your feet are closer to the hole than your head, so the hole pulls on them harder than on your head. The same was true, of course, when you used to stand on the Earth; but the head-to-foot difference on Earth was so minuscule, less than one part in a million, that you never noticed it. By contrast, as you float in your capsule at a circumference of 100,000 kilometers, the head-to-foot difference is one-eighth of an Earth gravity (⅛ "*g*"). At the center of your body the centrifugal force of your orbital motion precisely counteracts the hole's pull. It is as though gravity did not exist; you float freely. But at your feet, the stronger gravity pulls down with an added 1/16 *g*, and at your head the weaker gravity allows the centrifugal force to push up with an added 1/16 *g*.

Bemused, you continue your inward spiral; but your bemusement quickly changes to worry. As your orbit grows smaller, the forces on your head and feet grow larger. At a circumference of 80,000 kilometers the difference is a ¼-*g* stretching force; at 50,000 kilometers it is a full Earth gravity stretch; at 30,000 kilometers it is 4 Earth gravities. Gritting your teeth in pain as your head and feet are pulled apart, you continue on in to 20,000 kilometers and a 15-g stretching force. More than this you cannot stand! You try to solve the problem by rolling up into a tight ball so your head and feet will be closer together and the difference in forces smaller, but the forces are so strong that they will not let you roll up; they snap you back out into a radial, head-to-foot stretch. If your capsule spirals in much farther, your body will give way; you will be torn apart! There is no hope of reaching the horizon's vicinity.

Frustrated and in enormous pain, you halt your capsule's descent, turn it around, and start carefully, gently, blasting your way back up through circular, coasting orbits of larger and larger circumference and then into the belly of the starship.

Entering the captain's chamber, you vent your frustrations on the ship's master computer, DAWN. "Tikhii, tikhii," she says soothingly (drawing words from the ancient Russian language). "I know you are upset, but it is really your own fault. You were told about those head-to-foot forces in your training. Remember? They are the same forces as produce the tides on the oceans of the Earth."[22]

Thinking back to your training, you recall that the oceans on the side of the Earth nearest the Moon are pulled most strongly by the Moon's gravity